Shakir Showkat Sofi

Researcher in Tensor Methods and Machine Learning

Ph.D. candidate, ESAT-STADIUS, KU Leuven, Belgium · expected completion March 2027

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I develop tensor decomposition and tensor network methods to compress, complete, and model high-dimensional data, with applications in machine learning, spatiotemporal forecasting, and quantum state tomography.

Research Interests

- **Learning from incomplete data:** sparse and low-rank recovery from structured samples, with conditions for recoverability.
- **Quantum state tomography:** low-rank states, measurement design, tensor-network models, and DMRG-type algorithms.
- **Tensor methods in ML/DL:** system identification, model compression, tree-based models and spatiotemporal forecasting.

Education

Feb 2023 – Mar 2027 (exp.) | KU Leuven, ESAT-STADIUS, Leuven/Kortrijk, Belgium
Ph.D. in Engineering Science (Electrical Engineering)
Advisor: Prof. Lieven De Lathauwer Co-advisor: Prof. Fatemeh Mohammadi
- **Provisional title:** "Learning tensor networks from incomplete data"

Tensor decompositions · Tensor networks · Data compression and completion · Quantum state tomography

Sep 2020 – Jun 2022 | Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia
M.Sc. in Data Science **GPA:** 4.7/5
Advisor: Prof. Ivan Oseledets Co-advisor: Dr. Rukhsan Ul Haq
- **Thesis:** "Spatiotemporal forecasting with application to the weather forecast"

Matrix and tensor completion · Spatiotemporal forecasting · Machine learning

Aug 2015 – Aug 2019 | Islamic University of Science and Technology (IUST), Srinagar, J&K, India
B.Tech. in Electrical Engineering **GPA:** 9.18/10
Advisor: Dr. Shahkar Ahmad Nahvi
- **Thesis:** "Energy-based modeling of DC–DC power converters"

DC–DC power converters · Euler–Lagrange modelling · Simulation

Research Experience

Feb 2023 – Present | KU Leuven, ESAT-STADIUS, Kortrijk, Belgium
Doctoral Researcher
- Developing algebraic and optimisation-based algorithms for tensor completion from structured observations, with conditions for unique recovery.
- Applying tensor network models to low-rank quantum state tomography and, more generally, to compressed sensing-inspired problems.

2024 – Present | Flanders AI Research Program (FAIR), Flanders, Belgium
Researcher
- Developing tensor-based methods for sports analytics.
- Studying sensor placement for body-worn sensors, using structured sampling results from tensor completion to reconstruct joint data from a limited number of sensors.

2025 – Present | Tensor Decompositions and Decision Trees, KU Leuven, Belgium
Research Project
- Collaborating with the KU Leuven ITEC group on personalised course recommendation, contributing the matrix and tensor decomposition components.
- Exploring how tree-based learning can be combined with tensor decompositions for scalability (ongoing, early stage).

2023 – 2025 | pyTensorlab, KU Leuven, Belgium
Software Development Contributor
- Contributed to pyTensorlab, a Python package for tensor computations.
- Worked on tensorization and visualisation functionality and prepared demos for users.

Summer 2021 | **Skoltech & Tensor Field, Moscow, Russia**
Research Intern
- Worked on data-driven weather forecasting and spatiotemporal time-series analysis.
- This work fed into the M.Sc. thesis and the later *GeoInformatica* article.

Publications

Journal Articles

[J4] **S. S. Sofi** and L. De Lathauwer, "Tensor train completion from fiberwise observations along a single mode," *Mathematics*, vol. 14, no. 5, art. 922, MDPI, 2026. doi:10.3390/math14050922

[J3] N. Tokcan, **S. S. Sofi**, V. T. Pham, C. Prévost, S. Kharbech, B. Magnier, T. P. Nguyen, Y. Zniyed, and L. De Lathauwer, "Tensor decompositions for signal processing: Theory, advances, and applications," *Signal Processing*, vol. 238, art. 110191, Elsevier, 2026. doi:10.1016/j.sigpro.2025.110191

[J2] **S. S. Sofi** and I. Oseledets, "A case study of spatiotemporal forecasting techniques for weather forecasting," *GeoInformatica*, vol. 29, no. 2, pp. 275–298, Springer, 2025 (online Sep. 2024). doi:10.1007/s10707-024-00530-y

[J1] M. U. Munieeb, **S. S. Sofi**, F. Bashir, M. U. Hassan, and S. A. Nahvi, "Some contributions to Lagrangian modelling of power converters," *Franklin Open*, vol. 8, art. 100125, Elsevier, 2024. doi:10.1016/j.fraope.2024.100125

Peer-Reviewed Conference Proceedings

[C4] **S. S. Sofi**, C. Vermeylen, and L. De Lathauwer, "Tomography by design: An algebraic approach to low-rank quantum states," in *Proc. 34th European Signal Processing Conference (EUSIPCO)*, Bruges, Belgium, Sep. 2026, pp. 2521–2525. arXiv:2602.15202

[C3] **S. S. Sofi**, C. Vermeylen, and L. De Lathauwer, "Tensor train quantum state tomography using compressed sensing," in *Proc. 33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, Sep. 2025, pp. 1332–1336. 10.23919/EUSIPCO63237.2025.11226412

[C2] **S. S. Sofi**, S. Hendrikx, and L. De Lathauwer, "Tensor train completion of multi-way data observed along one mode," in *Proc. 32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, Aug. 2024, pp. 1067–1071. doi:10.23919/EUSIPCO63174.2024.10715267

[C1] **S. S. Sofi** and N. Alsahanova, "Image segmentation with topological priors," in *Proc. IEEE High Performance Extreme Computing Conference (HPEC)*, Sep. 2023, pp. 1–6. doi:10.1109/HPEC58863.2023.10363528

Preprints (under-review)

[P1] **S. S. Sofi**, C. Vermeylen, F. Mohammadi, and L. De Lathauwer, "A block tensor train Burer–Monteiro framework for low-rank quantum state tomography," arXiv:2609.09457 [quant-ph], 2026. arXiv

Talks and Posters

Invited and Seminar Talks

Oct 2026 "Learning tensor trains from incomplete data with applications to multiway data completion and quantum state tomography," Physics Seminars, KULAK, KU Leuven, Kortrijk, Belgium.

Sep 2026 "Learning tensor networks for positive semidefinite matrices from incomplete data," *Viksit Bharat 2047: Emerging Trends in Science and Technology, Economy and Business for Sustainable Development*, Srinagar, J&K, India.

Oct 2024 "Tensor completion methods," Flanders AI Research (FAIR) Day, Wintercircus, Ghent, Belgium.

Conference Talks

Sep 2025 "Tensor train quantum state tomography using compressed sensing," 33rd European Signal Processing Conference (EUSIPCO 2025), Isola delle Femmine – Palermo, Italy.

Jun 2025 "Tensor train completion of multiway data observed in a single mode and quantum state tomography (QST)," 26th Conference of the International Linear Algebra Society (ILAS 2025), NSYSU, Kaohsiung, Taiwan.

Aug 2024 "Tensor train completion of multi-way data observed along one mode," 32nd European Signal Processing Conference (EUSIPCO 2024), Lyon, France.

Sep 2023 "Image segmentation with topological priors," IEEE High Performance Extreme Computing Conference (HPEC 2023), online.

Posters

Sep 2026 "Tomography by design: An algebraic approach to low-rank quantum states," 34th European Signal Processing Conference (EUSIPCO 2026), Bruges, Belgium.

Jan 2026 "Tensor completion in TT format from structured fiber samples," 11th Workshop on Matrix Equations and Tensor Techniques (METT XI), ESAT, KU Leuven, Belgium.

Jan 2026 "Block tensor train approach for compressed sensing quantum state tomography," iBOF Study Day: Tensor Tools for Taming the Curse, ESAT, KU Leuven, Belgium.

Teaching

Apr 2024 – Present | **KU Leuven, ESAT, Bruges, Belgium**
Instructor – *Applied AI: Academic Perspectives (AI & Tensors)*, advanced master's course
- Lectures on tensor methods in AI learning: from PCA to multiway PCA; denoising and completion with the multilinear SVD (MLSVD); the canonical polyadic decomposition (CPD) and its applications; and an introduction to tensor networks.

Mar 2023 – Present | **KU Leuven, ESAT, Campus Kulak Kortrijk, Belgium**
Teaching Assistant – *Systems Theory and Control Engineering*, undergraduate course
- Prepare and run theory and programming exercise sessions on discrete- and continuous-time systems, integral transforms, sampling and reconstruction, and discretisation.
- Develop assignments and assist with assessment and grading.

KIMS – Quantum AI Lab, Srinagar, J&K, India
Instructor – *Python Programming: Foundations and Applications*
- Taught a module on linear algebra and its applications in Python.

Schools and Training

Jun 2026 EECI International Graduate School on Control, *Deep Learning for System Identification*, KU Leuven, Belgium.

Summer 2021 QWorld (Global Quantum Network) Summer School, quantum computing and programming with Qiskit, virtual.

Scholarships

2023 Doctoral scholarship, KU Leuven, Belgium.

2020 Graduate fellowship, Skolkovo Institute of Science and Technology, Moscow, Russia.

2015–2019 Merit-cum-Means Scholarship, Ministry of Human Resource Development (MHRD), Government of India.

Peer Review

Journals IEEE Transactions on Signal Processing; IEEE Signal Processing Letters; IEEE Transactions on Quantum Engineering; *Automatica* (Elsevier); *Natural Hazards* (Springer).

Conferences European Signal Processing Conference (EUSIPCO); IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP).

Professional Memberships

- Society for Industrial and Applied Mathematics (SIAM)
- European Association for Signal Processing (EURASIP)

Technical Skills

Tensor methods CPD for pattern identification and interpretation; MLSVD and tensor trains for compressing higher-order data; low-rank matrix and tensor completion (e.g., climate data).

ML & DL Supervised and unsupervised learning; deep learning for sequential and spatiotemporal modelling; meta-learning; neural network compression via tensor decompositions.

Quantum Low-rank quantum state tomography; tensor-network-based state preparation; compressed sensing.

Programming Python (NumPy, PyTorch, Xarray, Darts, pyTensorlab); MATLAB (Tensorlab, Tensor Toolbox, TT-Toolbox); C; SQL; LaTeX.

Languages English, Urdu, Kashmiri.

References

Available upon request.